L04: Learning in games

Foundations

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Can equilibrium arise from natural dynamics of learning agents?

It depends on what "arise" means—and whether the agents are good enough.

However: the answer is surprisingly positive

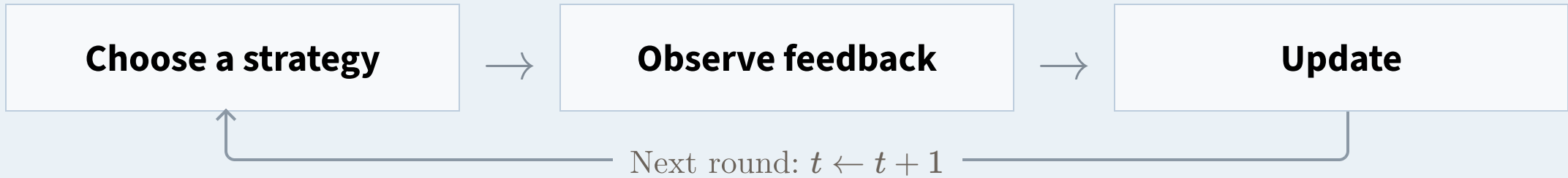
Suitable notions of equilibrium can be extracted efficiently, no matter the number of players, actions, whether the game is simultaneous or sequential, etc.

This is despite the fact that multiagent learning can give rise to formally *chaotic* dynamics.

How do we formalize learning?

1. Interaction model

At every round $t = 1, 2, \ldots$



2. Quality metric: How do we measure if a "learner is learning"?

Consider that everything is nonstationary: no *fixed* objective to optimize once and for all...

Interaction model: what can the agent do and see?

Several choices to make:

Strategy

One action

$a^{(t)} \in A$

or

*** Distribution**

$\boldsymbol{x}^{(t)} \in \Delta(A)$

$\rightarrow$

Feedback

Expected utility of the played distribution

Single real number

or

*** Utility of every action (counterfactuals)**

Entire loss function; equivalently, vector of values for each of the actions the learner can pick

*** Full-information model**

The simplest model; directly tied to **convex optimization**; the basis for more complicated settings.

Good news: sampling and utility estimation reduce the other models to this core.

The full-information model: Notation

$\mathcal{X} \subseteq \mathbb{R}^d$	**Strategy set** Nonempty, convex, compact; for normal-form games, $\mathcal{X} = \Delta(A)$.
$\boldsymbol{x}^{(t)} \in \mathcal{X}$	**Our choice at round t** Chosen using the feedback from earlier rounds.
$u^{(t)} : \mathcal{X} \to \mathbb{R}$	**Utility function at round t** Our utility is $u^{(t)}(\boldsymbol{x}^{(t)})$.

Full information: observe the whole utility function

Choose $\boldsymbol{x}^{(t)}$, then observe $u^{(t)}$, then update for round $t+1$.

Normal-form games: linear utility

In normal-form games the utility is linear in the learner's strategy

$$u^{(t)}(\boldsymbol{x}) = \sum_{a \in A} x_a g_a^{(t)} = \langle \boldsymbol{g}^{(t)}, \boldsymbol{x} \rangle$$

Example (RPS): $\mathcal{X} = \Delta(\{R, P, S\})$. Our distribution is $\boldsymbol{x}^{(t)}$; the opponent's is $\boldsymbol{y}^{(t)}$.

Our payoff matrix U

	✊	✋	✌
✊	0	-1	$+1$
✋	$+1$	0	-1
✌	-1	$+1$	0

What feedback is revealed?

$$u^{(t)}(\boldsymbol{x}) := \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}^{(t)}$$

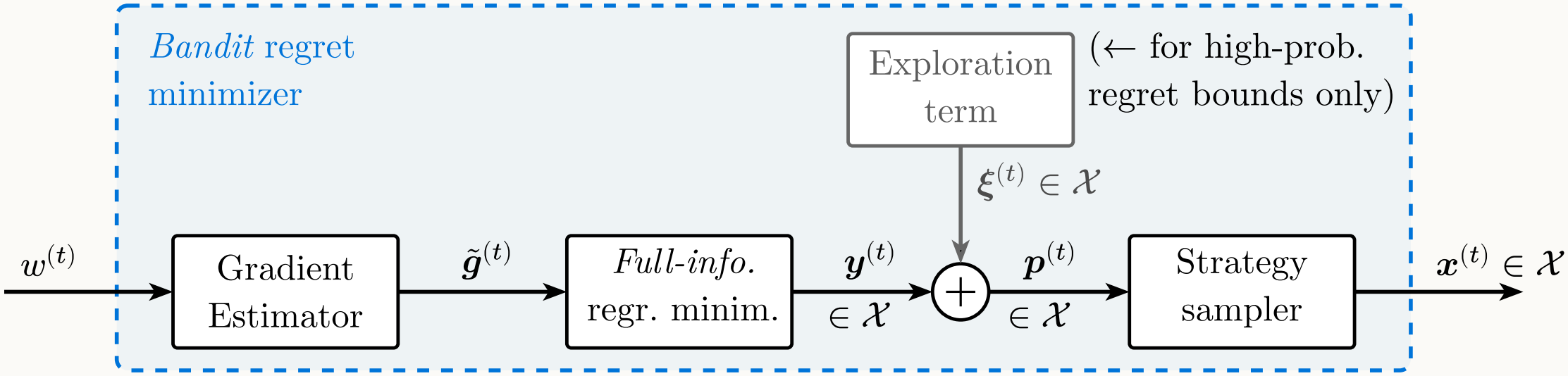
We give $\boldsymbol{u}^{(t)}$ via its gradient:

$$\boldsymbol{g}^{(t)} := U\boldsymbol{y}^{(t)} = \begin{pmatrix} u(\text{rock}, \boldsymbol{y}) \\ u(\text{paper}, \boldsymbol{y}) \\ u(\text{scissors}, \boldsymbol{y}) \end{pmatrix}$$

Bandit feedback? (L06)

Bandit feedback: observe only the utility of what we played

One number $w^{(t)} = u^{(t)}(\boldsymbol{x}^{(t)})$; the utilities of counterfactual alternatives are hidden.

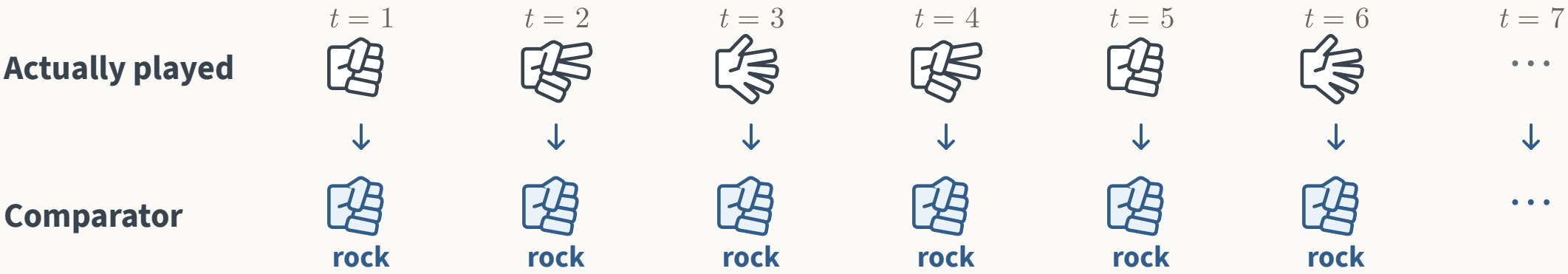


Idea: Estimate a utility vector from the observed utility, then feed the estimate to the learner.

The notion of regret

Consider this scenario:

Can you say you learned if, looking back at the history of play, you wish you could throw out everything you've done, and stick to one time-independent strategy instead?



Regret

$$\mathrm{Reg}^{(T)} := \max_{\hat{x} \in \mathcal{X}} \sum_{t=1}^{T} \left[u^{(t)}(\hat{x}) - u^{(t)}(\boldsymbol{x}^{(t)}) \right]$$

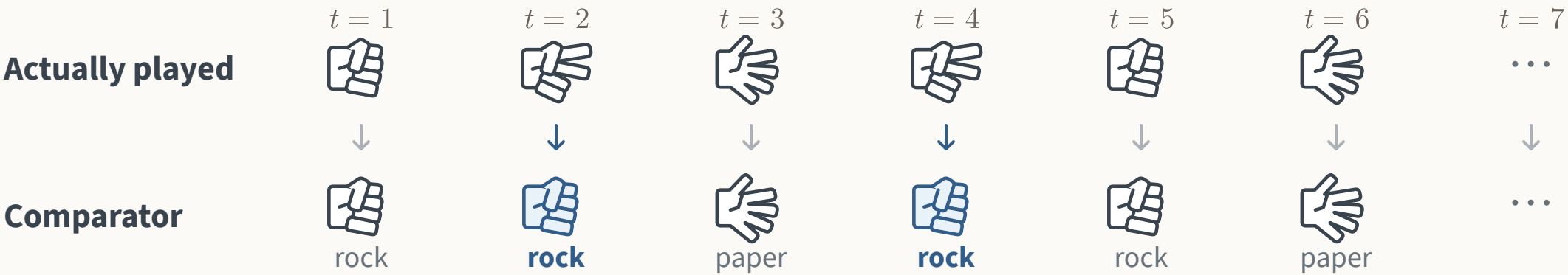
should not grow *too large* (to be defined later...)

Φ-Regret

External regret is a very basic desideratum: a **bare minimum**.

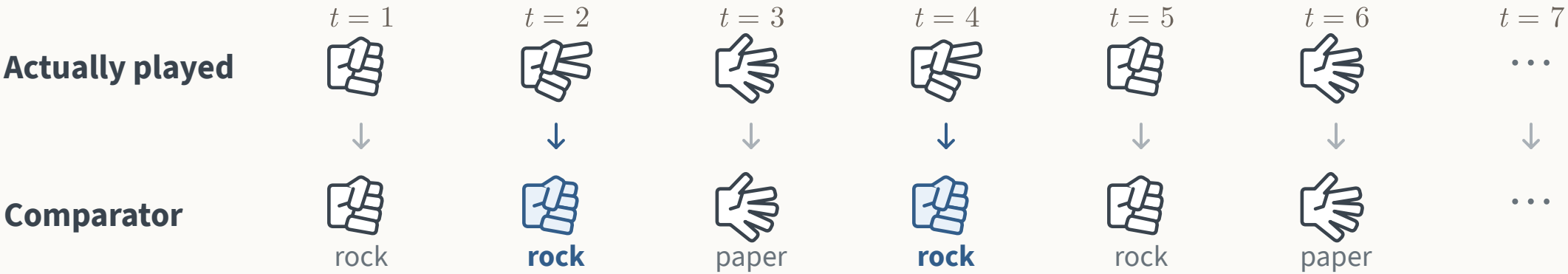
"We should not wish to go back and throw out everything."

How about more fine-grained changes?



Can you say you learned if, looking back at the history of play, you wish you could apply one deterministic transformation ϕ to everything you've done, and do substantially better?

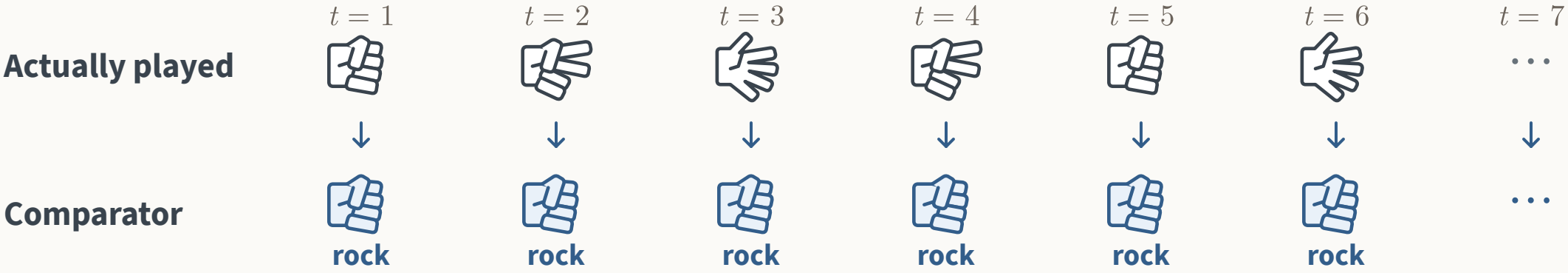
A framework: Φ-regret



$$\Phi\text{-}\operatorname{Reg}^{(T)} := \max_{\phi \in \Phi} \sum_{t=1}^{T} \left[u^{(t)}\left(\phi(\boldsymbol{x}^{(t)})\right) - u^{(t)}(\boldsymbol{x}^{(t)}) \right]$$

External regret

Constant replacement: discard the input strategy. Here, always rock.



$$\Phi^{\text{ext}} = \{\phi_{\hat{\boldsymbol{x}}} : \boldsymbol{x} \mapsto \hat{\boldsymbol{x}} \mid \hat{\boldsymbol{x}} \in \Delta(A)\}$$

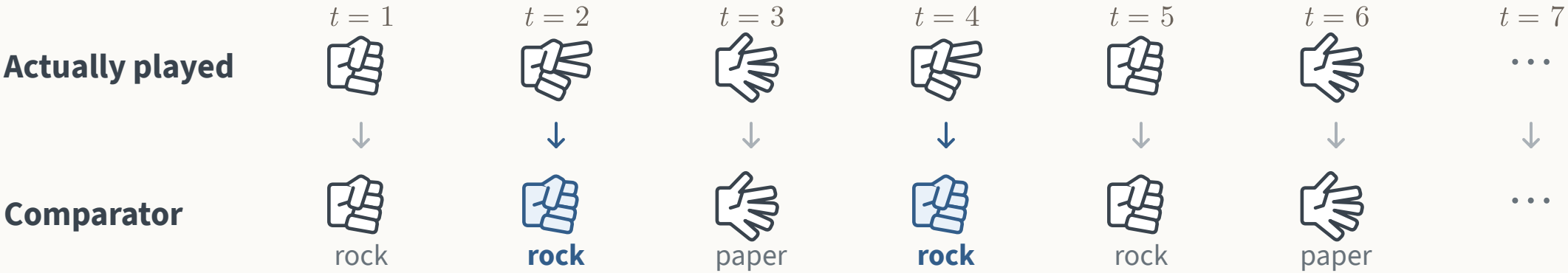
Theorem · Equilibrium approximation

Average joint play is an ε-**CCE**, with $\varepsilon = \max_i \dfrac{\text{Reg}_i^{(T)}}{T}$.

Two-player zero-sum: average strategies have **Nash gap** $\dfrac{\text{Reg}_1^{(T)} + \text{Reg}_2^{(T)}}{T}$.

Internal regret

One action-to-action switch: move all mass from a to b; leave the rest unchanged.



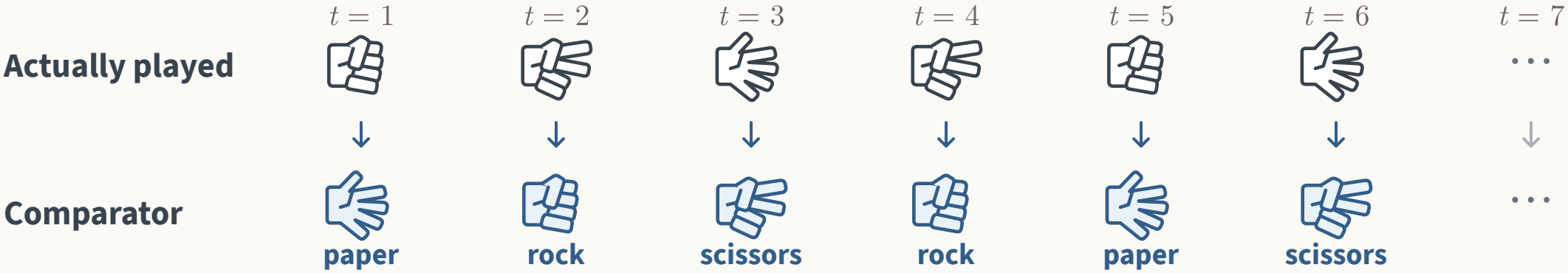
$$(\phi_{a\to b}(\boldsymbol{x}))_s = \begin{cases} 0 & s = a, \\ x_b + x_a & s = b, \\ x_s & \text{otherwise}, \end{cases} \qquad a \neq b$$

Theorem · Computing a correlated equilibrium

Average joint play is an ε-**CE**, with $\varepsilon = |A| \max_i \dfrac{\Phi\text{-}\operatorname{Reg}_i^{(T)}}{T}$.

Swap regret

A replacement for every action: allow all stochastic maps of the simplex.



$$\boxed{\Phi^{\text{swap}} = \{\boldsymbol{x} \mapsto Q\boldsymbol{x} : Q \geq 0,\ \mathbf{1}^\mathsf{T} Q = \mathbf{1}^\mathsf{T}\}}$$

I.e., Q is column-stochastic. Column a of Q gives the replacement distribution for action a.

Theorem · Computing a correlated equilibrium

Average joint play is an ε-**CE**, with $\varepsilon = \max_i \dfrac{\Phi\text{-}\operatorname{Reg}_i^{(T)}}{T}$.

Trigger deviations in sequential games

Connections between regret and equilibria go well beyond normal-form games.

Example: Sequential games.

Φ = Follow recommendations until the trigger; then switch to a fixed continuation plan.

Trigger regret → extensive-form correlated equilibrium

Average correlated play is an ε-**EFCE**, with $\varepsilon = |A| \max_i \frac{\Phi\text{-}\operatorname{Reg}_i^{(T)}}{T}$.

Sublinear regret is enough

In light of the previous connections,

The goal of a "regret minimizer"

Any nontrivial regret guarantee is already interesting. (Say, even $T^{0.99}$ or $T/\log T$ is enough asymptotically.)

A natural goal is then to make sure that

$$\boxed{R_i^{(T)} = o(T)} \quad \text{that is,} \qquad \max_i \frac{R_i^{(T)}}{T} \longrightarrow 0$$

The surprising might of external regret

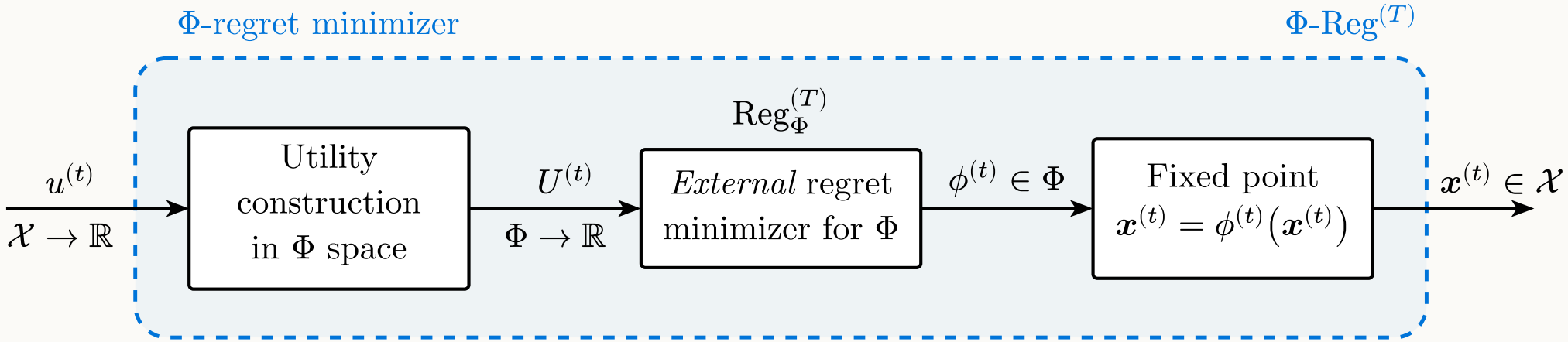
From external regret to Φ-regret

External regret "feels" like such a weak notion of rationality: just make sure you stay competitive with throwing out everything and use a fixed strategy...

Surprising fact

General Φ-regret can be reduced to external regret (albeit on a more complex space) by a beautiful construction of Gordon, Greenwald, and Marks (2008).

Idea: Embed an external regret minimizer **on** Φ and a **fixed-point oracle**.



$$U^{(t)}(\phi) := u^{(t)}(\phi(x^{(t)}))$$

The minimax theorem

Cool fact!

The existence of no-regret algorithms is enough to prove the minimax theorem (under the hypotheses we saw last time). Bonus: it proves it **constructively**. It teaches us how to construct an offense strategy from defense strategies.

$$\max_{\boldsymbol{x}\in\mathcal{X}} \min_{\boldsymbol{y}\in\mathcal{Y}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y} = \min_{\boldsymbol{y}\in\mathcal{Y}} \max_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}$$

Let a regret minimizer play against an opponent who best responds each round.

Part I: Weak duality

Let

The $\boldsymbol{x}$-player plays **offense**

$$\boxed{V_- := \max_{\boldsymbol{x}\in\mathcal{X}} \min_{\boldsymbol{y}\in\mathcal{Y}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}}$$

1. $\boldsymbol{x}$ chooses $\rightarrow$ 2. $\boldsymbol{y}$ responds

The $\boldsymbol{x}$-player plays **defense**

$$\boxed{V_+ := \min_{\boldsymbol{y}\in\mathcal{Y}} \max_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}}$$

1. $\boldsymbol{y}$ chooses $\rightarrow$ 2. $\boldsymbol{x}$ responds

von Neumann's minimax theorem guarantees $V_- = V_+$.

1. The direction $V_+ \geq V_-$ is free. No smart construction is needed. *Why?*

2. **Idea**: In V_+, the x-player can always reuse the commitment that guarantees V_-.

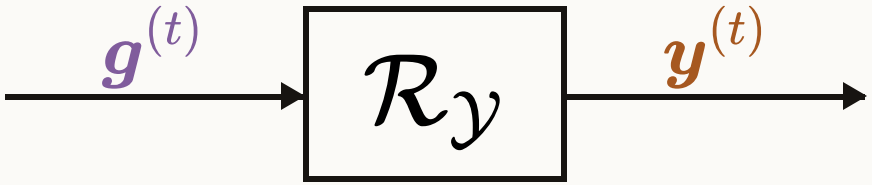
3. Let $\boldsymbol{x}^*$ be a maximizer in V_-. Then

$$V_- = \min_{\boldsymbol{y}\in\mathcal{Y}} \boldsymbol{x}^{*\mathsf{T}} U \boldsymbol{y}$$

$$V_+ = \min_{\boldsymbol{y}\in\mathcal{Y}} \max_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y} \geq \min_{\boldsymbol{y}\in\mathcal{Y}} \boldsymbol{x}^{*\mathsf{T}} U \boldsymbol{y} = V_-$$

Part II: Strong duality via regret

Interesting direction (**strong** duality): $\boxed{V_- := \max_{\boldsymbol{x}\in\mathcal{X}} \min_{\boldsymbol{y}\in\mathcal{Y}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}} \geq \boxed{V_+ := \min_{\boldsymbol{y}\in\mathcal{Y}} \max_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}}.$



$$\boldsymbol{x}^{(t)} \in \arg\max_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}^{(t)}$$
$$\boldsymbol{g}^{(t)} := -U^\mathsf{T} \boldsymbol{x}^{(t)}$$

Claim: $\bar{\boldsymbol{x}} := \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}^{(t)}$ is an **offense** strategy (i.e., approximate solution to V_-), and satisfies

$$V_- \geq \boxed{\min_{\boldsymbol{y}\in\mathcal{Y}} \bar{\boldsymbol{x}}^\mathsf{T} U \boldsymbol{y} \geq V_+ - \frac{\mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T}}.$$

I. The no-regret $\mathcal{Y}$-learner receives utility $u^{(t)}(\boldsymbol{y}) = \langle \boldsymbol{g}^{(t)}, \boldsymbol{y}\rangle = -\boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}$. Hence,

$$\mathrm{Reg}_{\mathcal{Y}}^{(T)} = \sum^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}^{(t)} - \min \sum^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}$$

Part II: Strong duality via regret

$$V_+ := \min_{\boldsymbol{y}\in\mathcal{Y}} \max_{\boldsymbol{x}\in\mathcal{X}} \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}$$

$$V_- \geq \boxed{\min_{\boldsymbol{y}\in\mathcal{Y}} \bar{\boldsymbol{x}}^\mathsf{T} U \boldsymbol{y} \geq V_+ - \frac{\mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T}}.$$

I. The no-regret $\mathcal{Y}$-learner receives utility $u^{(t)}(\boldsymbol{y}) = \langle \boldsymbol{g}^{(t)}, \boldsymbol{y}\rangle = -\boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}$. Hence,

$$\mathrm{Reg}_{\mathcal{Y}}^{(T)} = \sum_{t=1}^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}^{(t)} - \min_{\boldsymbol{y}\in\mathcal{Y}} \sum_{t=1}^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}$$

II. Divide by T and rearrange:

$$\min_{\boldsymbol{y}\in\mathcal{Y}} \bar{\boldsymbol{x}}^\mathsf{T} U \boldsymbol{y} = \boxed{\frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}^{(t)} - \frac{\mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T} \geq V_+ - \frac{\mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T}}.$$

Concluding the proof. The last step comes from $\boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}^{(t)} \geq V_+$. *Why is that true?*

Hence: Strong duality $V_- \geq V_+ - \mathrm{Reg}_{\mathcal{Y}}^{(T)}/T$. Letting $T \to +\infty$, we conclude $V_- = V_+$.

External regret learns best responses

A basic sanity check: when the environment is static, learning should exploit it. (Multiagent learning extends single-agent learning)

Against *stationary* stochastic opponents, a no-external-regret learner guarantees that

$$\overline{\boldsymbol{x}}_i^{(T)} := \frac{1}{T} \sum_{t=1}^{T} \boldsymbol{x}_i^{(t)}$$

is a best response.

Proof sketch

$$\mathrm{Reg}^{(T)} = \max_{\boldsymbol{x} \in \mathcal{X}} \sum_{t=1}^{T} u^{(t)}(\boldsymbol{x}) - \sum_{t=1}^{T} u^{(t)}(\boldsymbol{x}^{(t)})$$

Idea of the proof

If opponents play according to $\bar{\boldsymbol{x}}_{-i}$, $u(\boldsymbol{x}) := u_i(\boldsymbol{x}, \bar{\boldsymbol{x}}_{-i})$.

$$\max_{\boldsymbol{x}} \frac{1}{T} \sum_{t} u^{(t)}(\boldsymbol{x}) = \max_{\boldsymbol{x}} u_i(\boldsymbol{x}, \bar{\boldsymbol{x}}_{-i})$$

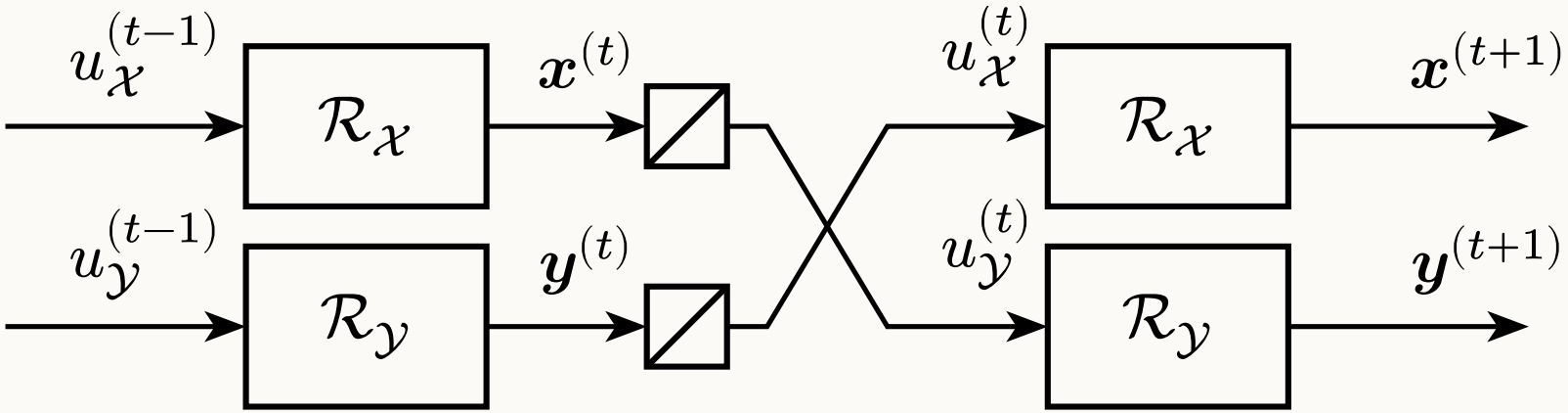
$$\frac{1}{T} \sum_{t} u^{(t)}(\boldsymbol{x}^{(t)}) = \frac{1}{T} \sum_{t} u(\boldsymbol{x}^{(t)}, \bar{\boldsymbol{x}}_{-i}) = u(\bar{\boldsymbol{x}}^{(T)}, \bar{\boldsymbol{x}}_{-i})$$

So:

$$0 \leftarrow \frac{\mathrm{Reg}^{(T)}}{T} = \max_{\boldsymbol{x}} u_i(\boldsymbol{x}, \bar{\boldsymbol{x}}_{-i}) - u(\bar{\boldsymbol{x}}^{(T)}, \bar{\boldsymbol{x}}_{-i})$$

Zero-sum self-play

$$u_1(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}, \qquad u_2 = -u_1$$



$$\boldsymbol{g}_{\mathcal{X}}^{(t)} = U \boldsymbol{y}^{(t)}, \qquad \boldsymbol{g}_{\mathcal{Y}}^{(t)} = -U^\mathsf{T} \boldsymbol{x}^{(t)}$$

The Nash gap of average play

$$\gamma(\overline{\boldsymbol{x}}, \overline{\boldsymbol{y}}) := \left(\max_{\hat{\boldsymbol{x}}} \hat{\boldsymbol{x}}^\mathsf{T} U \overline{\boldsymbol{y}} - \overline{\boldsymbol{x}}^\mathsf{T} U \overline{\boldsymbol{y}} \right) + \left(\overline{\boldsymbol{x}}^\mathsf{T} U \overline{\boldsymbol{y}} - \min_{\hat{\boldsymbol{y}}} \overline{\boldsymbol{x}}^\mathsf{T} U \hat{\boldsymbol{y}} \right) = \frac{\mathrm{Reg}_{\mathcal{X}}^{(T)} + \mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T}$$

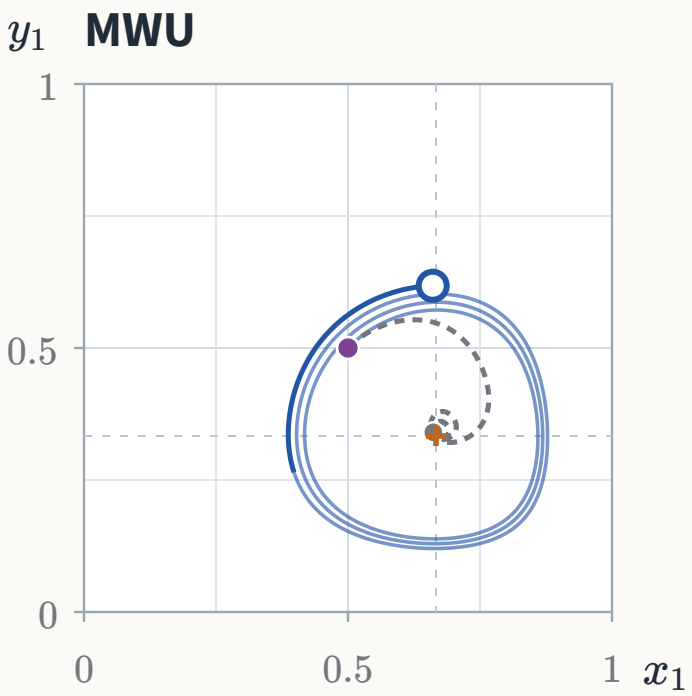
γ is the sum of unilateral deviation gains; $\gamma = 0$ exactly at Nash.

Any no-regret learners: convergence of averages

Blanket guarantee for averages

Any pair of no-regret learners works.

Remark: the theorem refers to the average of strategies, not the last strategy.



$$U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

Start

Last iterate

Nash $(\frac{2}{3}, \frac{1}{3})$

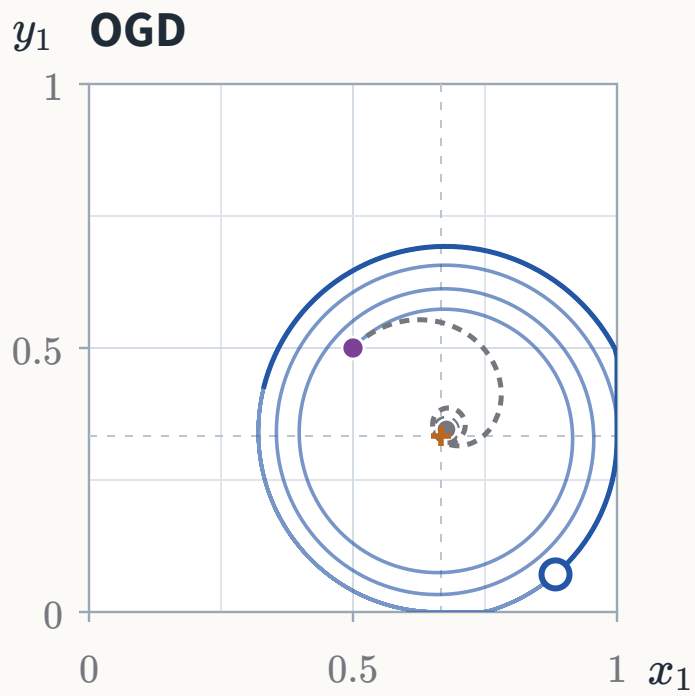
Average

Last-iterate gap

0.297

Average gap

0.017



$\eta = 0.0316$

$T_{\max} = 1{,}000$

Start

Last iterate

Nash $(\frac{2}{3}, \frac{1}{3})$

Average

Last-iterate gap

0.741

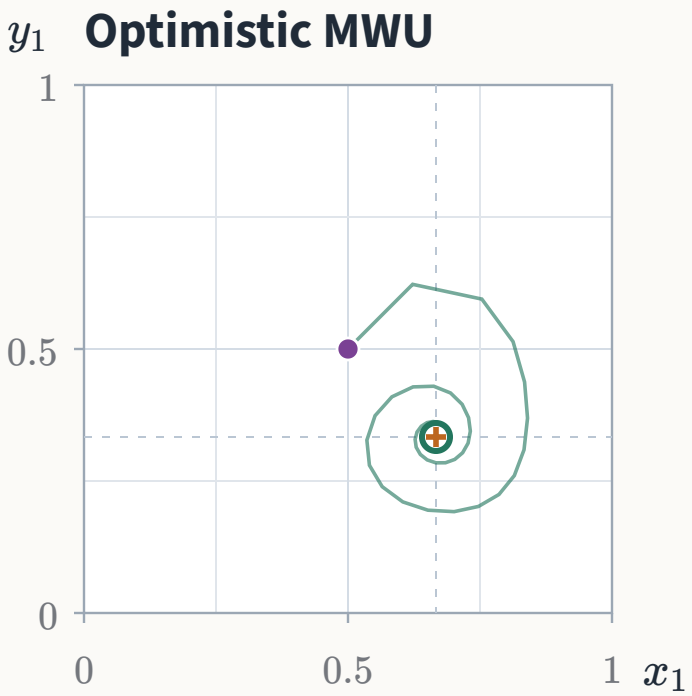
Average gap

0.023

Last-iterate convergence

With more structure, stronger conclusions

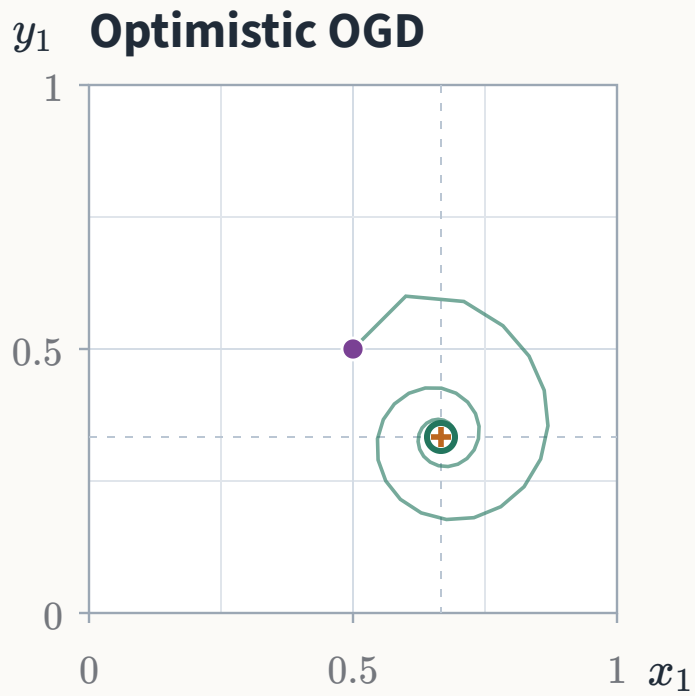
Suitable optimistic dynamics also make the *current strategies* converge in zero-sum games. (See **S04**)



$$U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

- Start
- Last iterate
- Nash $(\frac{2}{3}, \frac{1}{3})$

Last-iterate gap $< 10^{-12}$



$\eta_{\mathrm{MWU}} = 0.5$

$\eta_{\mathrm{OGD}} = 0.2$

- Start
- Last iterate
- Nash $(\frac{2}{3}, \frac{1}{3})$

Last-iterate gap $< 10^{-12}$

Proof sketch

$$\begin{aligned}\gamma(\boldsymbol{x}, \boldsymbol{y}) &= \left(\max_{\hat{\boldsymbol{x}}} \hat{\boldsymbol{x}}^\mathsf{T} U \boldsymbol{y} - \boldsymbol{x}^\mathsf{T} U \boldsymbol{y}\right) + \left(\boldsymbol{x}^\mathsf{T} U \boldsymbol{y} - \min_{\hat{\boldsymbol{y}}} \boldsymbol{x}^\mathsf{T} U \hat{\boldsymbol{y}}\right) \\ &= \max_{\hat{\boldsymbol{x}}} \hat{\boldsymbol{x}}^\mathsf{T} U \boldsymbol{y} - \min_{\hat{\boldsymbol{y}}} \boldsymbol{x}^\mathsf{T} U \hat{\boldsymbol{y}}\end{aligned}$$

Idea: write regret guarantees for the two players and sum

$$\frac{\mathrm{Reg}_{\mathcal{X}}^{(T)}}{T} = \max_{\hat{\boldsymbol{x}}} \hat{\boldsymbol{x}}^\mathsf{T} U \bar{\boldsymbol{y}} - \cancel{\frac{1}{T} \sum_{t=1}^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}^{(t)}}$$

$$\frac{\mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T} = \cancel{\frac{1}{T} \sum_{t=1}^{T} \boldsymbol{x}^{(t)\mathsf{T}} U \boldsymbol{y}^{(t)}} - \min_{\hat{\boldsymbol{y}}} \bar{\boldsymbol{x}}^\mathsf{T} U \hat{\boldsymbol{y}}$$

$$\boxed{\gamma(\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}) = \frac{\mathrm{Reg}_{\mathcal{X}}^{(T)} + \mathrm{Reg}_{\mathcal{Y}}^{(T)}}{T}}$$

Coarse correlated equilibrium

For any number of players and actions

$$\boldsymbol{\mu}^{(T)} = \frac{1}{T} \sum_{t=1}^{T} \boldsymbol{x}_1^{(t)} \otimes \cdots \otimes \boldsymbol{x}_n^{(t)}$$

$$\max_{\hat{a}_i} \; \mathbb{E}_{\boldsymbol{a} \sim \boldsymbol{\mu}^{(T)}} \left[u_i(\hat{a}_i, \boldsymbol{a}_{-i}) - u_i(\boldsymbol{a}) \right] = \frac{\mathrm{Reg}_i^{(T)}}{T}$$

External regret leads to coarse correlated equilibrium

Sublinear external regret for every player makes the average joint distribution approach CCE.